Statistics and Data Analysis

Week 3

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Program of the first 4 weeks (March)

Days	Thu 12:45 - 14:30 (2h)	Fri 13:45 - 15:30 (2h)
05/03 06/03	- Probability: foundations and interpretations - Practice with PDFs, CDFs, Quantiles - Joint PDFs, moments, correlation, transformations	- Practical CLT and the lognormal - Linear error propagation - MC methods - **Exercise**
12/03 13/03	- Frequentist hypothesis testing - Optimal tests, Neyman-Pearson, p-value - Estimators and properties	- Likelihood - MLE - **Exercise**
19/03 20/03	- Asymptotic MLE properties and MVB - Advanced MLE applications and MC methods - Frequentist Confidence intervals	- Profile likelihood and Wilks' theorem - **Exercise**
26/03 27/03	- Bayesian priors and posterior construction - Bayesian parameter estimation	- Bayesian model selection and evidence - Marginal likelihood, Bayes factor - **Exercise**





Jupyter notebook

Week2_Exercise_Solutions.ipynb

Recap[2]: Variance of estimators

We have seen several methods to estimate the variance of an estimator:

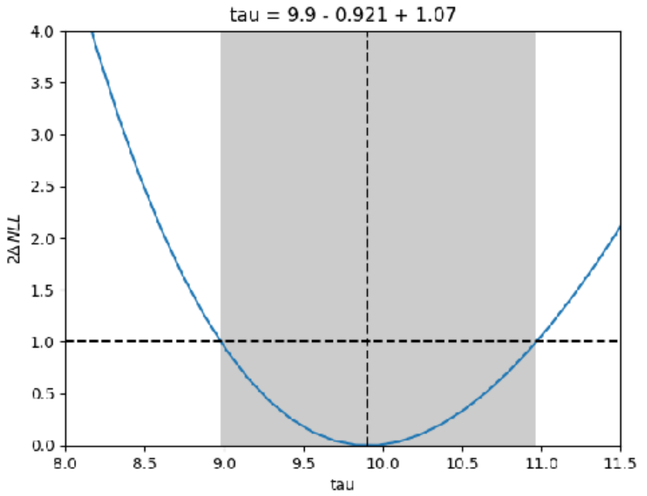
- **Analytic derivation** of $V[\hat{\theta}]$ for simple estimators (often impractical)
- **MC method**: simulate new datasets and calculate $\hat{\theta}$ estimates, get std deviation
  - If you don't have the true θ, just generate MC with $\hat{\theta}$ from the real dataset
- **Bootstrapping**: emulate new datasets using resampling with replacement
  - The sample at hand has to be representative of the underlying distribution
  - Can be bad with low stat and parameters related to the tails of the distribution

- **Minimum Variance Bound:**

$$\sigma^2_{\hat{\theta}} = \left(\frac{\partial^2 \mathrm{NLL}}{\partial \theta^2} \right)^{-1} \Bigg|_{\theta=\hat{\theta}}$$

- **Graphical method:**

$$2\Delta NLL = -2\left(\ln(L) - \ln(L_{\max})\right) = 1$$



…can get asymmetric errors (confidence interval defined later)

Information inequality (1 parameter)

- The **information inequality** (Minimum Variance Bound) for one parameter states that:

$$V[\hat{\theta}] \geq \frac{\left(1 + \frac{\partial b}{\partial \theta}\right)^2}{E\left[-\frac{\partial^2 \ln L}{\partial \theta^2}\right]}$$

Where b is the bias $b = E[\hat{\theta}] - \theta$ and the denominator is called “Fisher information”

- We have seen that a Maximum Likelihood Estimator (MLE) in the **asymptotic limit** of $n \to \infty$

$$\lim_{n\to\infty} V[\hat{\theta}] = -\frac{1}{E\left[\frac{\partial^2 \ln L}{\partial \theta^2}\right]}$$

And this formula calculated at $\theta = \hat{\theta}$ is used to get an estimate of the variance (e.g. in Minuit)

Information inequality (N parameters)

- For a PDF of $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_N)$ the **information inequality** states that:

  The matrix M_{ij} is positive semi-definite (i.e. $z^{\mathrm{T}} M z \geq 0$ for any $z \neq 0$)

$$M_{ij} = V_{ij} - \sum_{k,l} \left(\delta_{ik} + \frac{\partial b_i}{\partial \theta_k} \right) I^{-1}_{kl} \left(\delta_{lj} + \frac{\partial b_l}{\partial \theta_j} \right)$$

  and I_{ij} is the Fisher information:

$$I_{ij} = -E \left[\frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j} \right]$$

- In the **asymptotic limit** of $n \to \infty$

$$b = 0 \text{ and } M_{ij} = 0$$

$$\text{and therefore } V_{ij} = I^{-1}_{ij}$$

  - So we can get an estimate of the covariance matrix by calculating the Fisher information matrix I_{ij} and inverting it
  - That is the what Minuit (Hesse) and fitting algorithm will do for you numerically



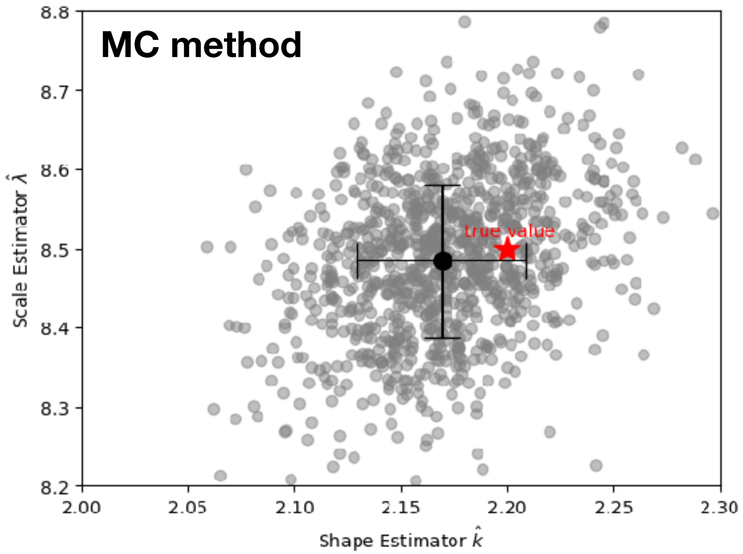


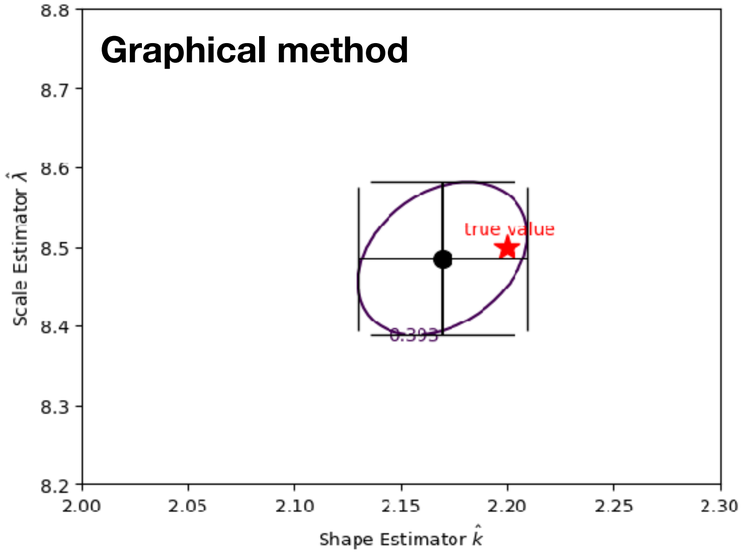
Jupyter notebook

Week3_MLE_Ndim.ipynb

Recap: Variance of N estimators

- How to estimate the covariance matrix for N estimators?
- **MC method or Bootstrapping**:
  - Generate pseudo-experiments
  - Calculate estimators
  - Get sample covariance
- **Information inequality**:
  - Valid in asymptotic limit
  - Calculate Fisher information matrix
  - Invert it (numerically) to get covariance
- **Graphical method**:
  - Scan likelihood and get contour $2\Delta LL = 1$
  - Variance from contour and ρ from tilt of ellipse





Frequentist confidence interval

Confidence interval by inverting a test

…also called "*Neyman construction*"

- **Goal**: report range of values for estimate to reflect stat uncertainty
  - We have seen how to estimate the standard deviation
  - Sometimes this is not sufficient

→ **frequentist confidence interval**

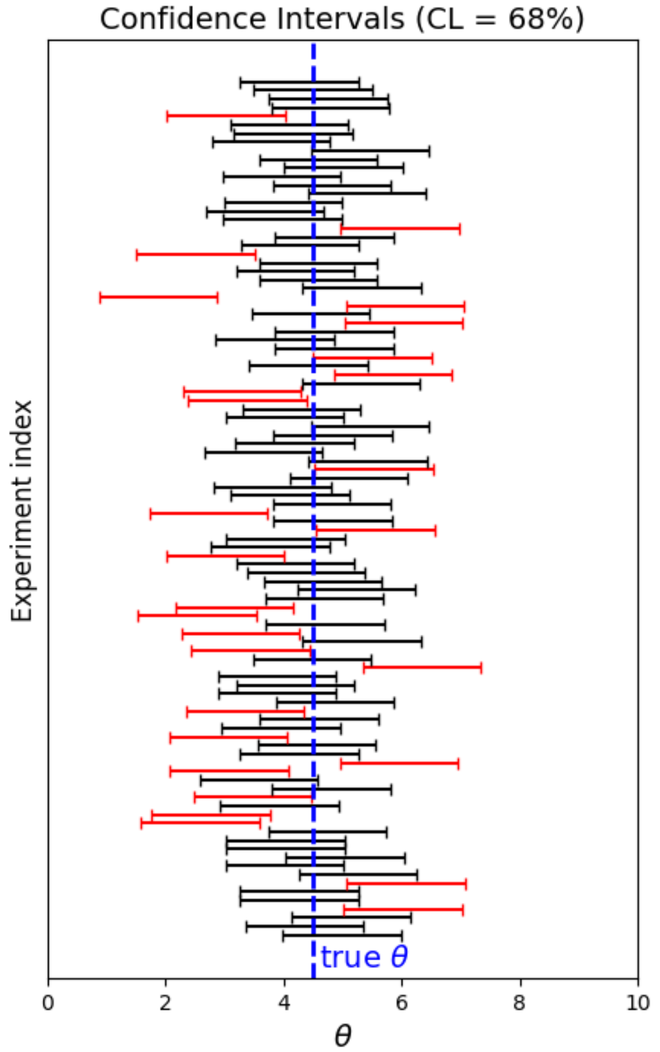
- The definition involves a hypothesis test and its p-value
  - Scan the values of the parameter θ (to test several hypotheses)
  - Calculate the p-value p_θ of the observed data for each of them
  - Confidence interval constituted by θ values that pass the test $p_\theta \geq \alpha$
- We call this construct a confidence interval at confidence level $1 - \alpha$

Coverage of confidence interval

- If the true value of θ is rejected, then it's not in the confidence interval
- The probability to reject the true value of θ will be α
- Therefore, the probability for the interval to "cover" (contain) the true value of θ is the confidence level

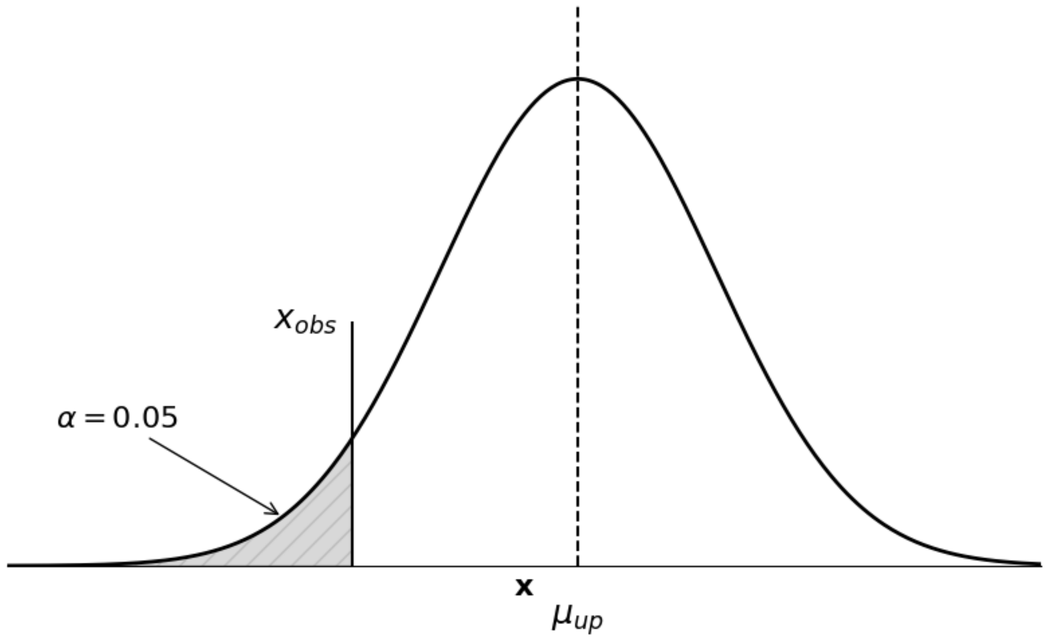
$$\mathrm{CL} = 1 - \alpha$$

- Frequentist interpretation with repeated experiments:
  - You repeat the experiment N times (sample PDF with true value θ)
  - You calculate a confidence interval on θ with CL=68% for each experiment
  - Then 68% of the confidence intervals will contain the true θ



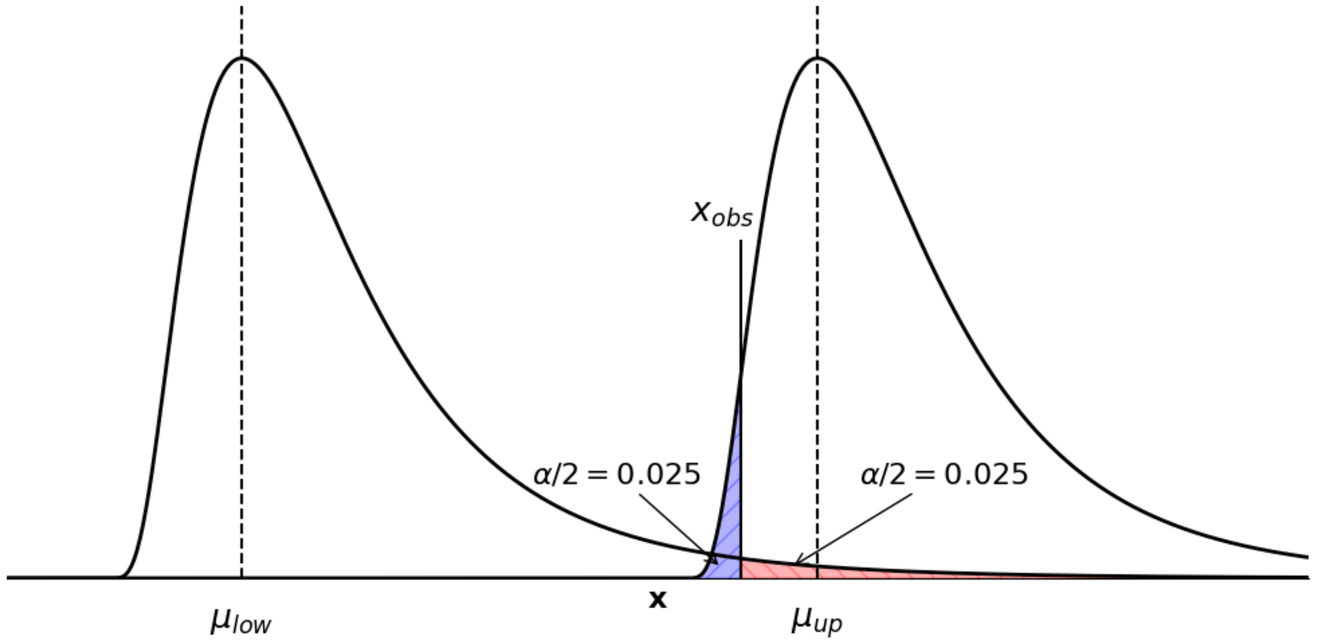
Example: upper limit on Gaussian μ

- Let's consider a single measurement x_{obs} of a variable that is known to follow a Gaussian distribution with unknown μ (and known σ)
- What's the upper limit μ_{up} at $\text{CL} = 1 - \alpha$ on the value of μ?
- μ_{up} will be the hypothetical value of μ such that under that hypothesis there is only a probability α to find $x < x_{\text{obs}}$



Two-sided confidence intervals

- Suppose a test where the alternative hypothesis includes both higher and lower values of μ (the critical region of the hypothesis test is arbitrary anyway)
- You can easily define a two-sided confidence interval that will have $\alpha/2$ probability of being too high and $\alpha/2$ of being too low



The confidence belt

Figures from G.Cowan chapter 9

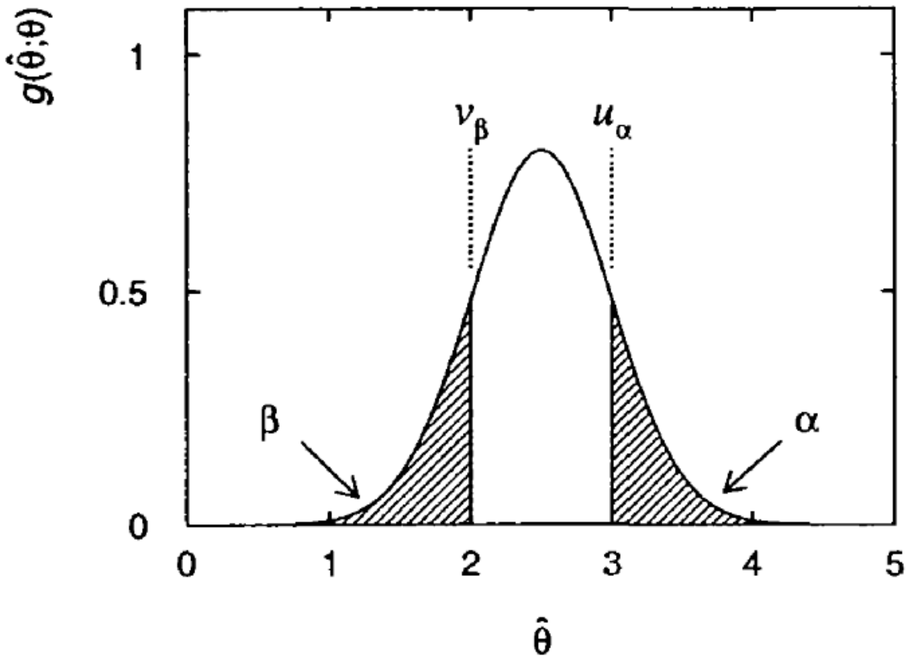


Fig. 9.1 A p.d.f. $g(\hat{\theta};\theta)$ for an estimator $\hat{\theta}$ for a given value of the true parameter θ. The two shaded regions indicate the values of $\hat{\theta} \leq v_\beta$, which has a probability β, and $\hat{\theta} \geq u_\alpha$, which has a probability α.

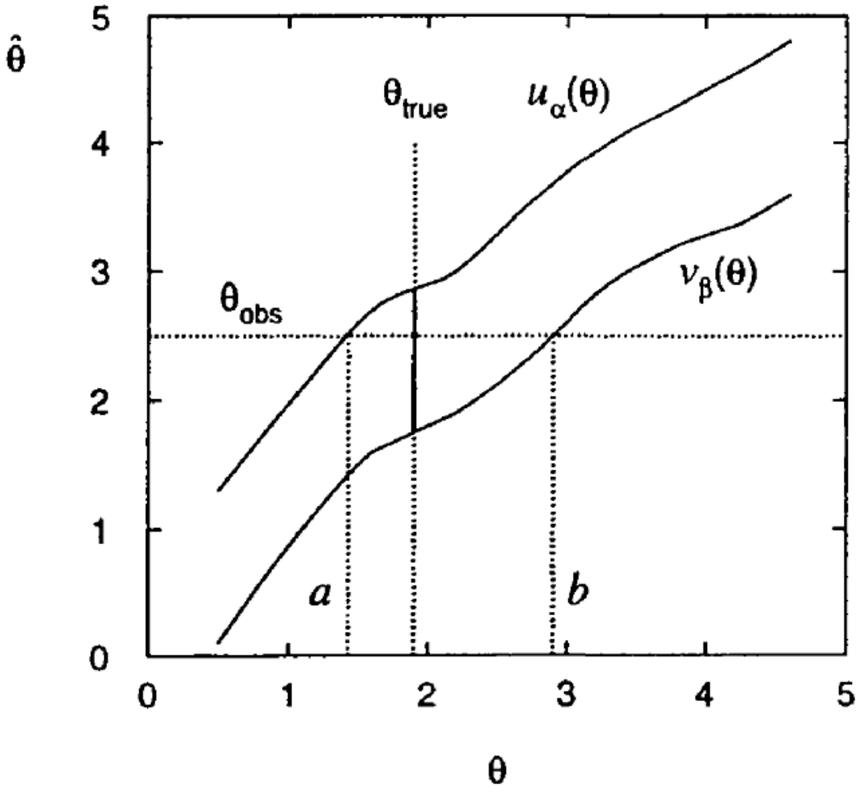


Fig. 9.2 Construction of the confidence interval $[a, b]$ given an observed value $\hat{\theta}_{obs}$ of the estimator $\hat{\theta}$ for the parameter θ (see text).

Meaning of frequentist confidence interval

- Often reported together with the “point estimate” of the parameter as asymmetric uncertainty:

$$\hat{\theta}^{+\sigma_{\text{up}}}_{-\sigma_{\text{low}}}$$

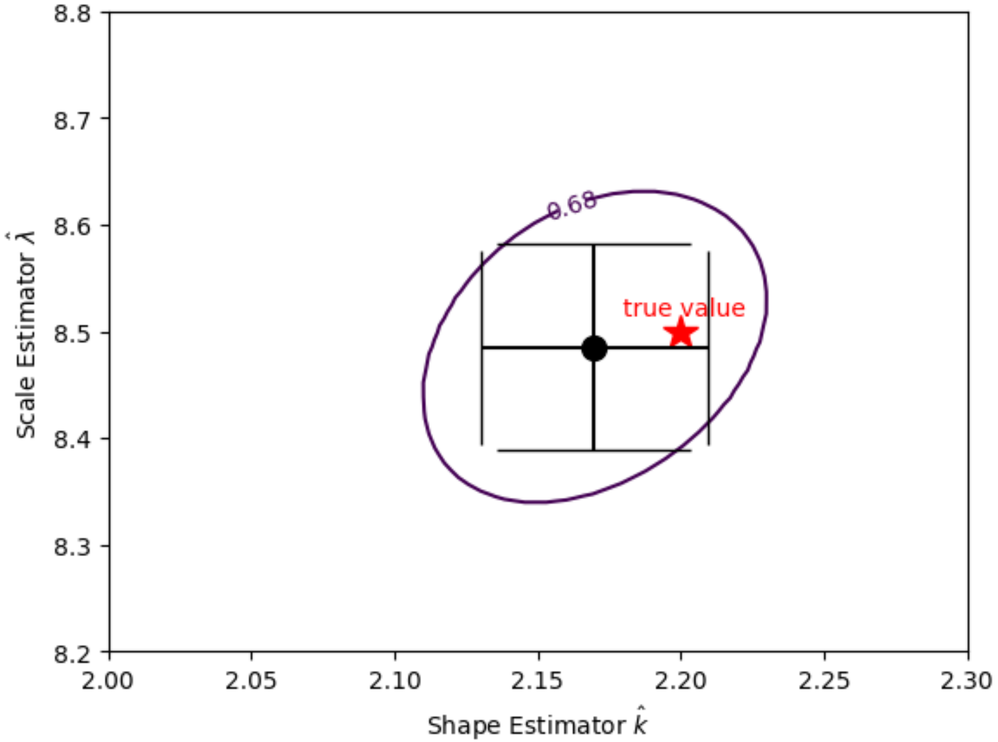
- is meant to represent a CL=68.3% confidence interval for θ of

$$\left[\hat{\theta} - \sigma_{\text{low}},\ \hat{\theta} + \sigma_{\text{up}}\right]$$

- **Typical misconception**: thinking of CL of the frequentist confidence interval as the probability that the parameter is in that range
- But the parameter θ does not fluctuate upon repetition of the measurement
- It is the interval that fluctuates upon repetition of the measurement
- the CL is the fraction of times you expect the interval to include the true θ

Confidence regions (multidimensional)

- In case we are dealing with N parameters we can define confidence regions in N-dimensional space
- Same recipe: set up a hypothesis test and invert it
- **Reminder**: Hypothesis testing with several parameters means defining a **test statistic** to get an ordering in the multidimensional parameter space



Confidence region from the Likelihood

- There is a recipe to get approximate confidence intervals/regions on a MLE $\hat{\theta}$ from the likelihood function
- We can define the test statistic as the likelihood ratio between the MLE and any other point in the parameter space $\boldsymbol{\theta}$:

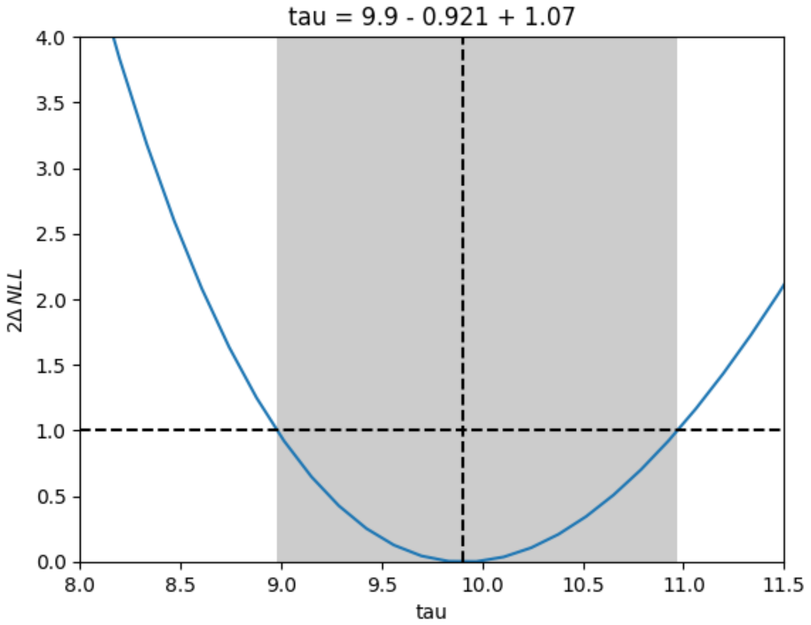
$$\lambda(\theta) = \frac{L(\theta)}{L(\hat{\theta})}$$

or taking the log: $t_\theta = 2\Delta NLL(\theta) = -2\left(\ln L(\theta) - \ln L(\hat{\theta})\right)$

  - So that higher t_θ means worse agreement between hypothesis θ and the data

- The p-value is then given by untegrating the test statistic PDF:

$$p_{\boldsymbol{\theta}} = \int_{t_{\boldsymbol{\theta},\,\mathrm{obs}}}^{\infty} f\left(t_{\boldsymbol{\theta}} \mid \boldsymbol{\theta}\right) dt_{\boldsymbol{\theta}}$$



but we don't have $f(t_\theta \mid \theta)$…

Wilks' theorem approximation

- Under certain regularity conditions and large-sample limit one can apply **Wilks' theorem** to get

  $f(t_\theta | \theta) \simeq \chi^2_n$ (the chi-square distr. with n dof)

  - The chi-square distribution with n being the number of parameters in $\boldsymbol{\theta}$

- Then to get the boundary you just set $p_\theta = \alpha$

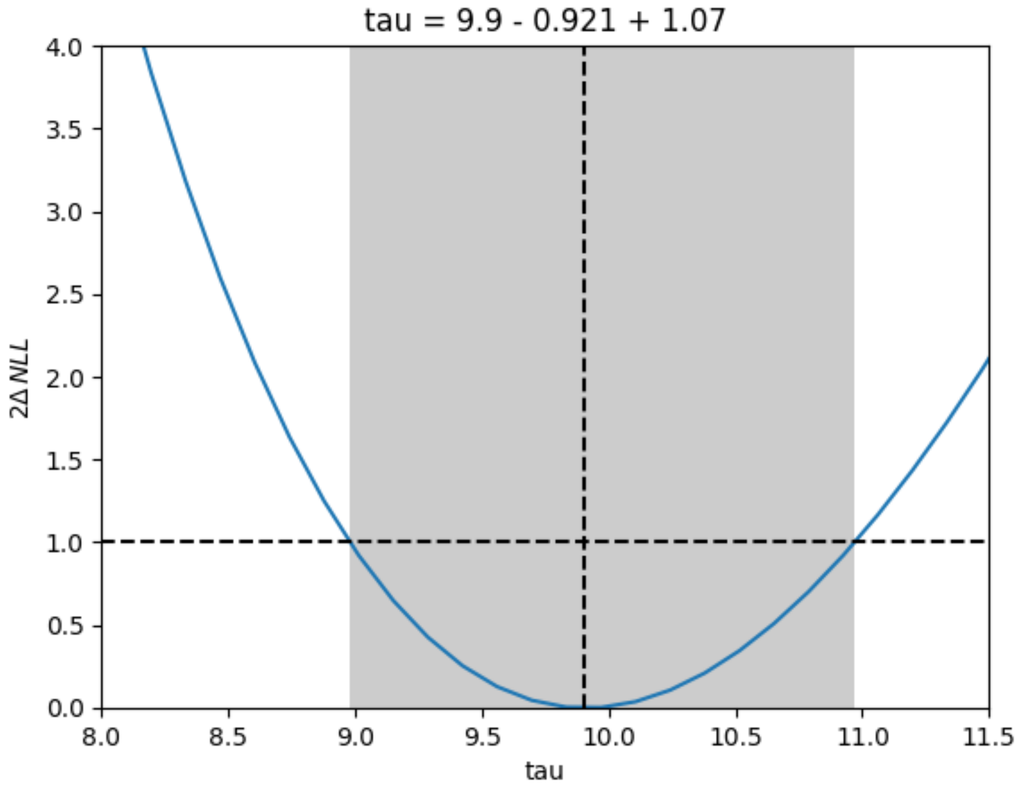
$$p_{\boldsymbol{\theta}} = 1 - F_{\chi^2_n}\left(t_{\boldsymbol{\theta}}\right) = \alpha$$

$$t_\theta = 2\Delta NLL(\theta) = F^{-1}_{\chi^2_n}(1-\alpha)$$

- And for example for $n = 1$ and $1 - \alpha = 68.3\,\%$ you get $F^{-1}_{\chi^2_n}(1-\alpha) = 1$

  → you can get a confidence interval by scanning $2\Delta NLL(\theta)$ like it's done by Minos (from Minuit)

The original paper from Wilks:
S.S. Wilks - Annals Math.Statist. 9 (1938) 1, 60-62

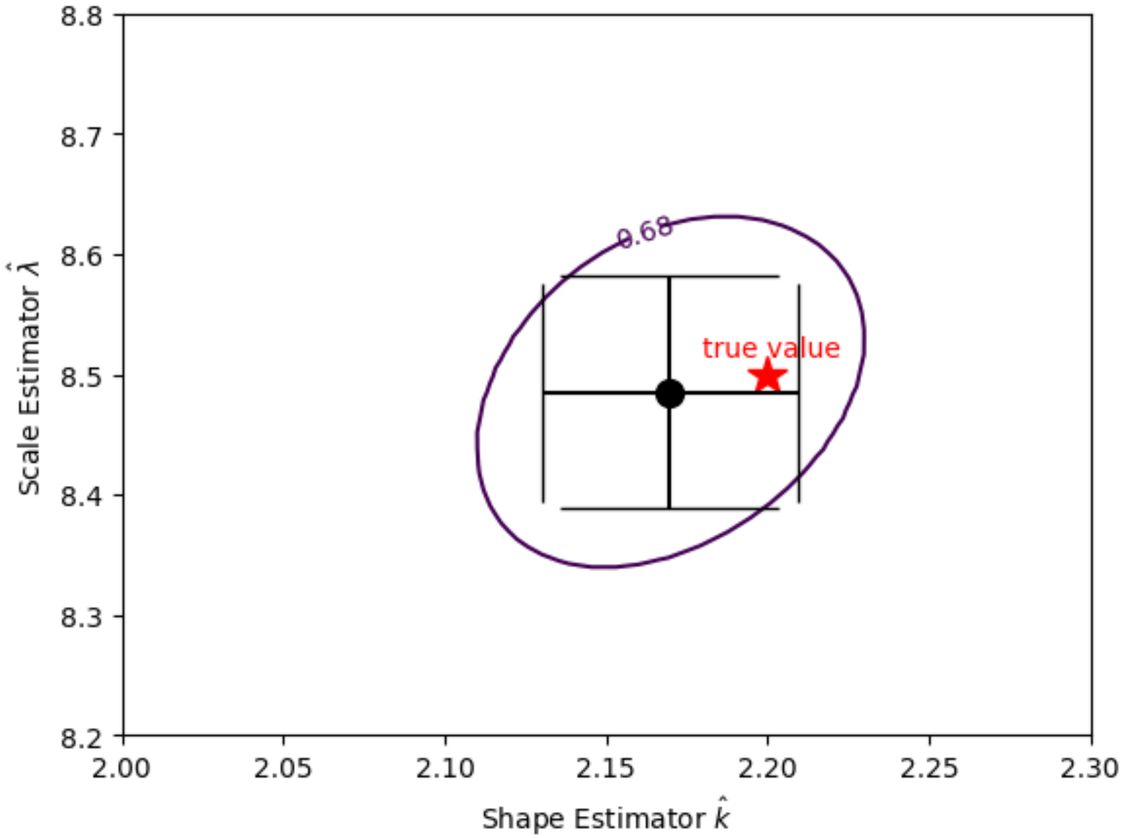


Example in two dimensions

- We already saw an example with $n = 2$
- In that case the test-statistic $2\Delta NLL(\theta)$ is predicted to follow a χ^2_n with $n = 2$
- Then the confidence region with $\mathrm{CL} = 1 - \alpha = 68.3\,\%$ is then given by

$$2\Delta NLL(\theta) = F^{-1}_{\chi^2_n}(1-\alpha) = 2.30$$

- Minos will automatically calculate $F^{-1}_{\chi^2_n}(1-\alpha)$ for you when you ask for the confidence region with a certain CL



Nuisance parameters and profile likelihood

Nuisance parameters and POI

- In realistic physics models, the full set of parameters $\boldsymbol{\theta}$ is rarely composed entirely of things we want to measure

- We can split model parameters in two categories $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\nu})$

  - Parameters of Interest $\boldsymbol{\mu}$: the (physical) quantities we want to measure (e.g. Higgs boson mass, yield of signal events, …)

  - Nuisance parameters $\boldsymbol{\nu}$: parameters we do not care about but must be included in the model to describe the data (e.g. background yield, detector resolution, calibration constants, …)

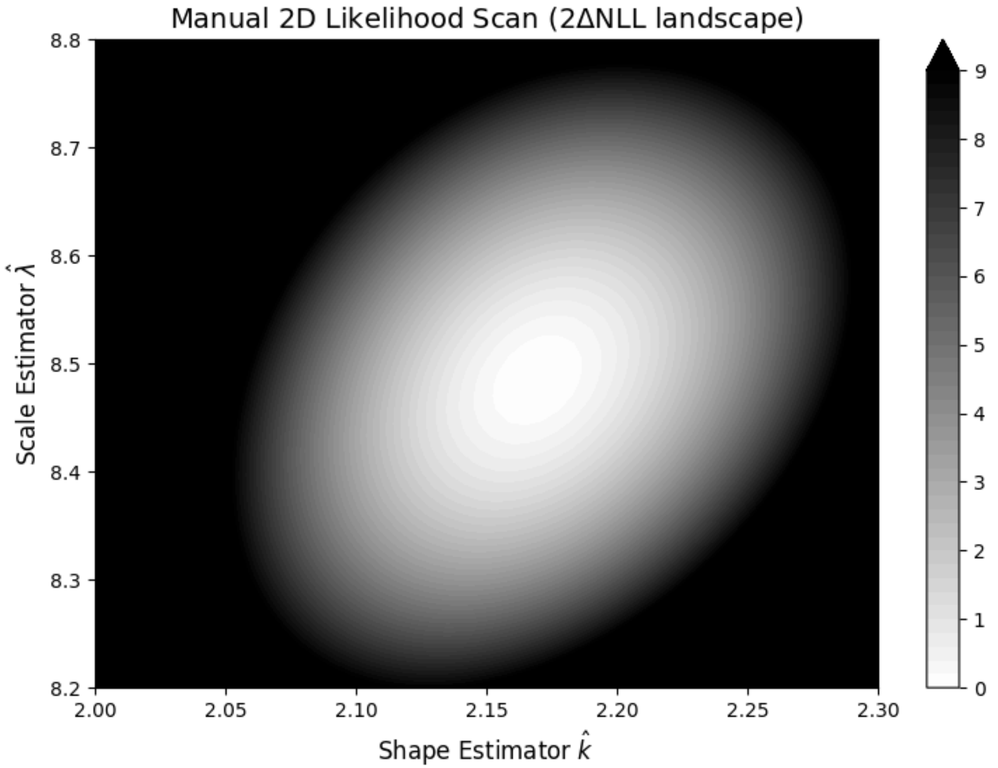
- How do we estimate the uncertainty (confidence region) on our POI while rigorously accounting for the fact that we don't know the nuisance parameters?

The profile likelihood

- In frequentist statistics we don't have a PDF of the parameters, so we can't just marginalise (= integrate out) the nuisance parameters
- To eliminate the nuisance parameters from our frequentist confidence intervals/regions, we use a technique called "profiling"
- We define the **profile likelihood ratio** as:

$$\lambda(\mu) = \frac{L(\mu, \hat{\hat{\nu}}(\mu))}{L(\hat{\mu}, \hat{\nu})}$$

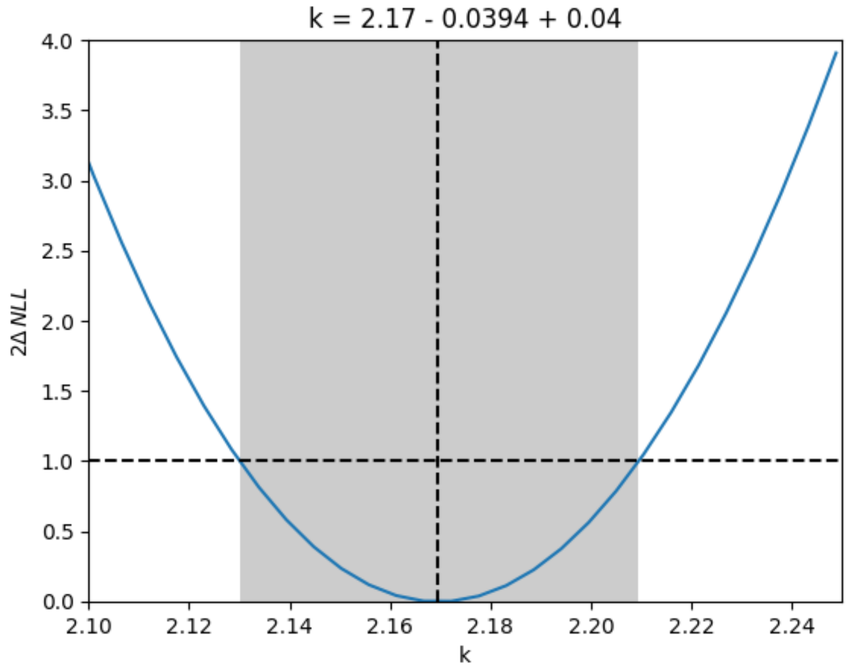
- *In practice*: we scan the values of μ and at every step we re-maximise the likelihood $L(\mu, \hat{\hat{\nu}}(\mu))$ with respect to the nuisance parameters



Wilks' theorem with profile likelihood

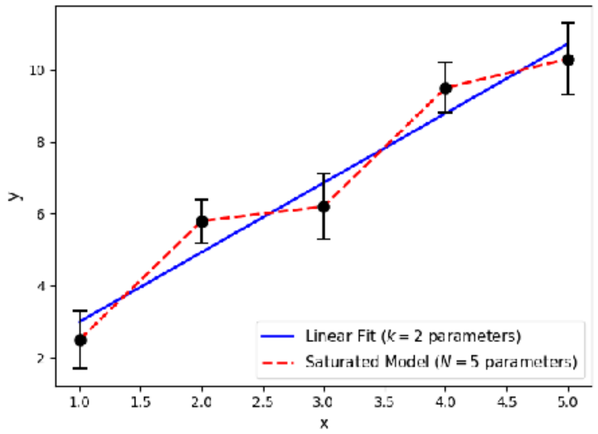
- Wilks' theorem ⇒ the test statistic obtained from the profile likelihood asymptotically follows a χ^2_n distribution with n degrees of freedom
- $n =$ (total # of parameters) − (# of nuisance parameters)
  $n =$ (# of POI)
- E.g. if you have just one POI, its **profile** likelihood will follow $\chi^2_{n=1}$ independently of the number of nuisance parameters
- You can use the profile likelihood as test statistic to define your confidence interval/region. In the large sample limit (and provided some regularity conditions) it will have good coverage

m.draw_mnprofile("k")



Goodness of fit and Wilks' theorem

- Supposed you have N independent measurements y_i at positions x_i and you fit a model $f(x;\boldsymbol{\theta})$ with K parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)$
- To test whether the model is a good description of the data we construct a likelihood ratio against a *saturated model*
- The saturated model perfectly matches data: $f_{\text{sat}}(x_i) = y_i$



- If we assume that we have Gaussian errors, the NLL reads:

$$-\ln L(\theta) = \text{const} + \sum_{i=1}^{N} \frac{\left(y_i - f\left(x_i;\theta\right)\right)^2}{2\sigma_i^2}$$

- But for the saturated model $y_i - f_{\text{sat}}(x_i) = 0$ so our test statistic becomes:

$$t_{\text{fit}} = \sum_{i=1}^{N} \frac{\left(y_i - f\left(x_i;\hat{\boldsymbol{\theta}}\right)\right)^2}{\sigma_i^2} \equiv \chi^2_{\text{min}}$$

- **Wilks' theorem** t_θ will follow a χ^2_n distribution with n = (pars in saturated) − (pars in model)
  $n = N - K$

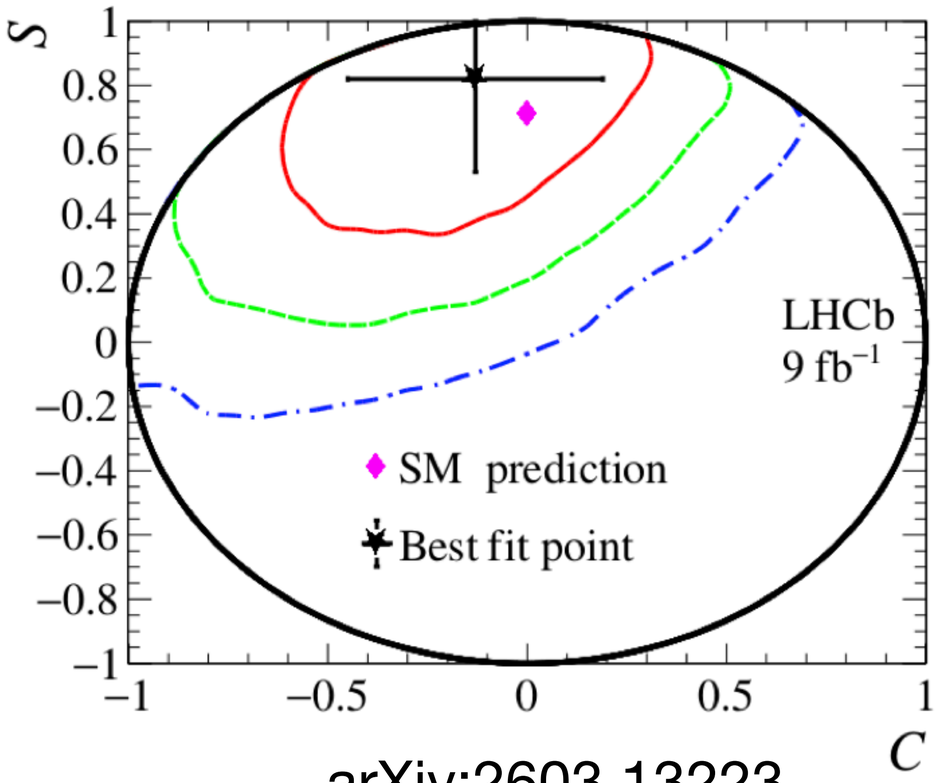
Some references for Wilks' theorem

1. The original paper: S.S. Wilks - Annals Math.Statist. 9 (1938) 1, 60-62
2. Sections 40.2.2.2, 40.3.2.1, 40.4.2.1 and 40.4.2.2 of the Statistics review of the Particle Data Group
3. The CCGV paper on asymptotic formulae for likelihood-based tests Eur.Phys.J.C 71 (2011) 1554

Example of 2D confidence region

First measurement of time-dependent CP violation in the flavor-changing neutral-current decay $B^0 \to K^0_S \mu^+ \mu^-$

Figure 4: Two-dimensional confidence regions for C and S based on a likelihood-ratio ordering technique, corresponding to the total q^2 range. The red solid, green dashed and blue dash-dotted contours represent the 68.3%, 95.5%, and 99.7% confidence intervals given the physical constraint $C^2+S^2 \leq 1$. The black marker with error bands represents the fit results for C and S. The magenta diamond marker stands for the SM prediction.



arXiv:2603.13223

Combining confidence intervals/regions

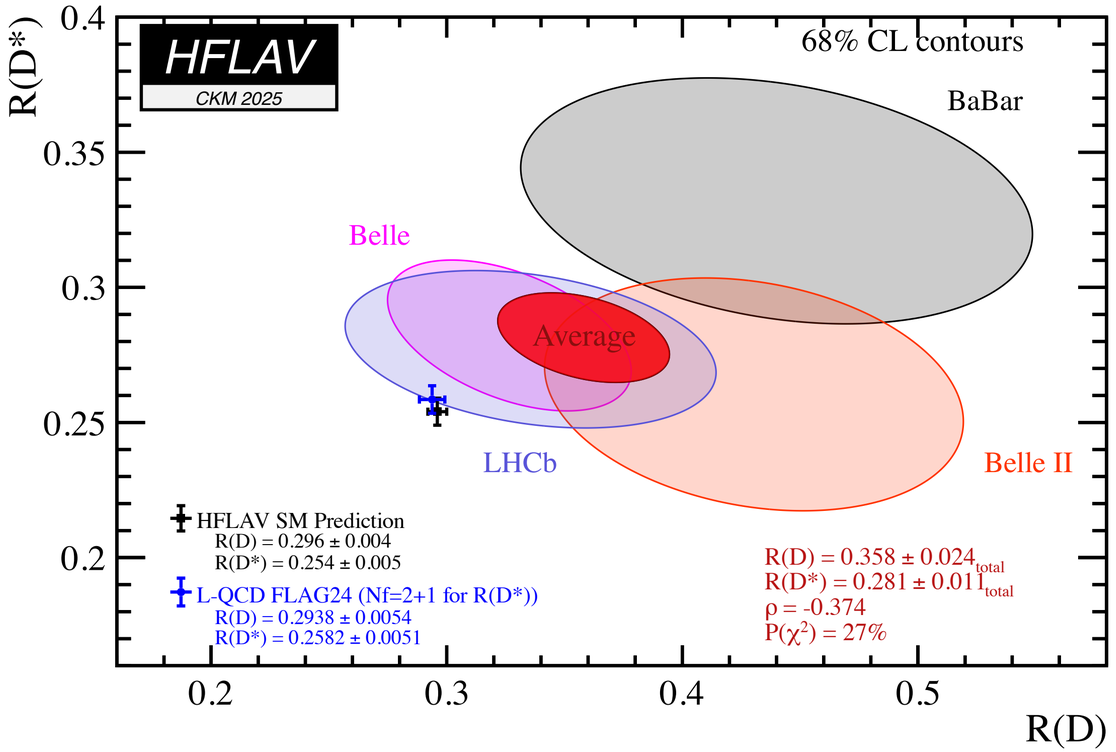
- Often one has N different experiments reporting a measurement of the same parameter(s) and one wants to combined them
- The best practice consists in combining the likelihoods

$$\ln L(\theta) = \sum_{i=1}^{N} \ln L_i(\theta)$$

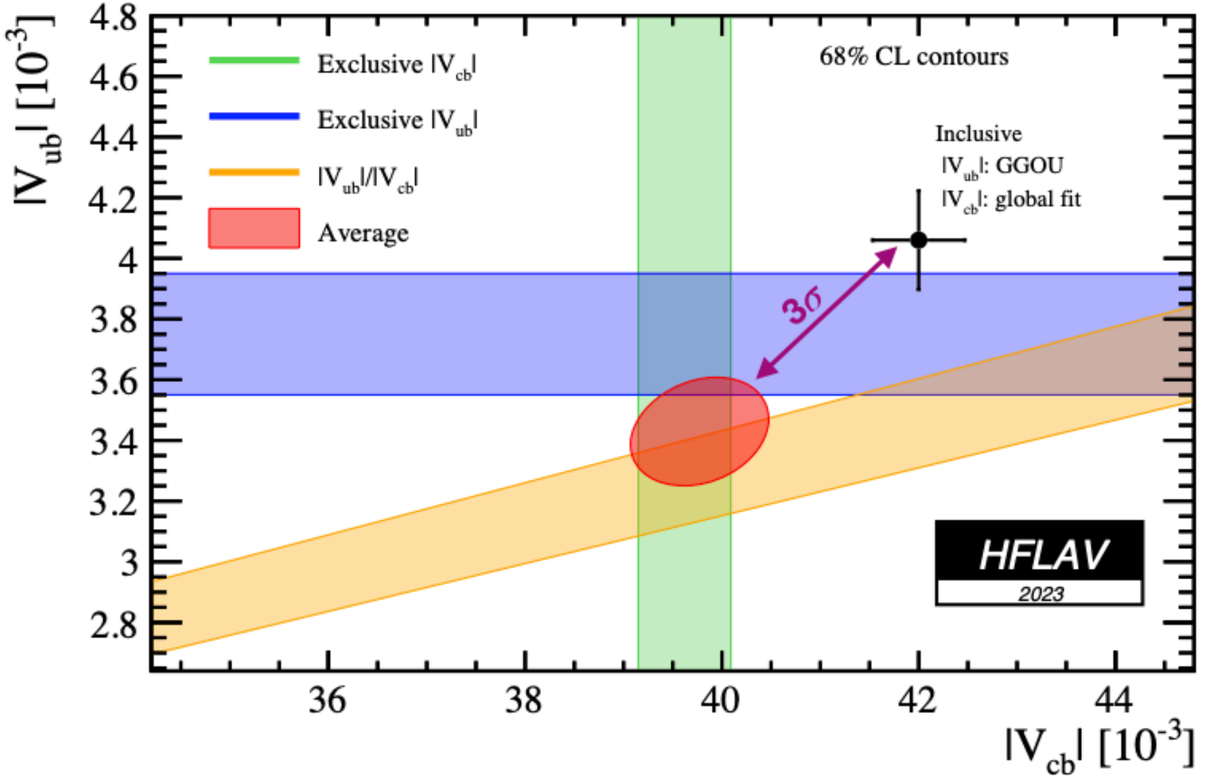
- It's good practice to provide the profiled likelihood with your measurement (e.g. https://www.hepdata.net/)
- If only the confidence interval/ region is provided one can reverse-engineer the likelihood
- Then one can calculate a combined confidence interval

Examples of likelihood combination

Tests of Lepton Flavour Universality



Quarks mixing matrix elements



https://hflav.web.cern.ch/

The frequentist toolbox



1. Hypothesis testing

- **Goal:** Choose between models (e.g., H_0: Background vs. H_1: Signal + Background).
- **Neyman-Pearson Lemma:** The Likelihood Ratio is the most powerful test statistic.
- **p-value:** The probability, under H_0, of observing data as extreme as ours or more.

2. Parameter Estimation (Finding the Best Value)

- **Goal:** Construct an estimator $\hat{\theta}$ with minimal **Bias**, maximal **Efficiency**, and **Consistency**.
- **Minimum Variance Bound (MVB):** sets the theoretical minimum on an estimator's variance.
- **Maximum Likelihood Estimator (MLE):** The ultimate workhorse. Asymptotically reaches the MVB (fully efficient) and is Gaussian distributed.
- **Fisher information** allows to estimate the covariance matrix from the second derivative of the Likelihood at $\theta = \hat{\theta}$

3. Confidence Intervals & Regions

- **Inverting a Hypothesis test:** guarantees exact frequentist **coverage** under repeated experiments.
- **Wilks' Theorem:** Asymptotically, $2\Delta\mathrm{NLL} \sim \chi^2_n$. It allows us to set confidence levels simply scanning the likelihood
- **Profile Likelihood:** The rigorous way to project a multi-dimensional likelihood onto the Parameters of Interest by fitting away **Nuisance Parameters**.

Exercise for week 3

Week3_exercise.ipynb

